Week 11 Self-Assessment

Introduction to Research Methods & DataLab

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1 Instructions

Test your understanding of this week's material. Answer each question to check your learning.

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2 Conceptual Questions

2.1 Question 1: Why Research Methods?

Why is "common sense" unreliable for understanding human behaviour?

- (A) It's always wrong
- (B) It suffers from biases like hindsight and confirmation bias
- (C) Scientists don't like common sense
- (D) It only works for simple behaviours

Show explanation Common sense is unreliable because of:

- **Hindsight bias** — events seem obvious after they occur
- **Confirmation bias** — we notice supporting evidence and ignore contradicting evidence
- **Anecdotal evidence** — individual cases can mislead
- Common sense can justify contradictory conclusions (e.g., "opposites attract" vs "birds of a feather flock together")

2.2 Question 2: Hallmarks of Science

```
opts_hallmarks <- c(
  "Popular, funded, published, and cited",
  answer = "Empirical, systematic, falsifiable, and transparent",
  "Theoretical, mathematical, complex, and reviewed",
  "Objective, simple, fast, and certain"
)
```

What are the four hallmarks of scientific claims?

- (A) Popular, funded, published, and cited
- (B) Empirical, systematic, falsifiable, and transparent

- (C) Theoretical, mathematical, complex, and reviewed
- (D) Objective, simple, fast, and certain

2.3 Question 3: Variable Types

A **continuous** variable can take any value within a range. TRUE / FALSE

A participant's **experimental condition** (A or B) is an example of a categorical variable. TRUE / FALSE

Reaction time in milliseconds is an example of a categorical variable. TRUE / FALSE

2.4 Question 4: Replication

What does it mean for a finding to be "replicable"?

- (A) The finding was published in a journal
- (B) The researcher repeated the study themselves
- (C) Other researchers get the same results following the same procedures
- (D) The finding agrees with common sense

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3 Practical Questions

3.1 Question 5: Measurement Error

In the reaction time task (ruler drop), which of the following could cause measurement error?

Inconsistent ruler height TRUE / FALSE

Participant knowing when the drop will happen TRUE / FALSE

Using a standardised ruler TRUE / FALSE

3.2 Question 6: Being a Participant and Tester

Why did you act as both participant AND tester in DataLab?

- (A) To save time
- (B) Because we didn't have enough researchers
- (C) To understand both perspectives and the importance of standardised procedures
- (D) To collect more data

3.3 Question 7: Signal Detection

In the staring task, what is a "hit"?

- (A) Saying 'not staring' when they were staring
- (B) Correctly detecting when someone IS staring
- (C) Saying 'staring' when they were NOT staring
- (D) Missing all trials

What is a "false alarm"?

- (A) Correctly detecting when someone is staring
- (B) Saying ‘not staring’ when they were staring
- (C) Saying ‘staring’ when they were NOT staring
- (D) Missing all trials

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4 Fill in the Blank

Complete these statements:

The four steps of the scientific method are: observe, ________, test, and conclude.

A variable that can only take whole number values (like “number of words recalled”) is called a ______ variable.

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5 Summary Checklist

Before moving on, check you understand:

- ☐ Why psychology uses research methods rather than relying on common sense
- ☐ The four hallmarks of scientific claims
- ☐ The difference between continuous, categorical, and discrete variables
- ☐ The steps of the scientific method
- ☐ What “replicable” means and why it matters